

When to Intervene: Comparing Constant and Physiology-Triggered Robotic Assistance

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Abstract

In human–robot interaction (HRI), deciding when to provide assistance requires understanding how intervention strategies affect both task performance and the user experience. Previous work compared physiology-triggered robotic assistance with periodic support in surgical training. We extend this work to physical spatial problem solving through a within-participant experiment comparing no assistance, constant assistance every 30 seconds, and adaptive assistance triggered by heart-rate signals. Twenty-four participants solve nine physical tangram puzzles across three condition blocks, with balanced condition orders and randomly assigned puzzles. Tangrams combine spatial reasoning with physical manipulation, providing a simple task for studying informational and physical assistance. Participants receive on-screen hints and robotic piece retrieval through a Wizard-of-Oz setup. We evaluate correct-piece progress, completion, round duration, adapted seven-point NASA-TLX workload, intervention experience, frustration, and final trust. Completion rates are 15.3% without assistance, 59.7% with constant assistance, and 50.0% with adaptive assistance. Both assisted conditions improve progress, completion, and workload relative to control (Holm-adjusted $p < .001$). Constant assistance has the highest observed completion and shortest duration; adaptive assistance has descriptively higher helpfulness and timing ratings and lower frustration, but no assisted-policy contrast survives multiplicity correction. These findings support assistance in physical spatial problem solving while leaving the choice between periodic and physiology-triggered assistance unresolved.

Keywords

Human–robot interaction, adaptive assistance, intervention timing, physiological sensing, Wizard-of-Oz, tangram

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1 Introduction

A pause during a physical puzzle can reflect productive mental rotation or uncertainty about the next move. A helpful intervention in the latter case may interrupt reasoning in the former. Assistance timing therefore matters alongside assistance content.



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Recent systems learn assistance type, timing, and confidence [1], estimate action completion for proactive coordination [10], or adapt task-level timing and role allocation [13]. Physiological information offers another timing cue: Yang et al. compared workload-adaptive robotic suction with periodic support in surgical training [33]. We extend that comparison to physical spatial problem solving, where text hints guide interpretation and an arm retrieves relevant pieces.

Tangrams combine spatial reasoning and manipulation using common materials, observable outcomes, and repeated targets; related HRI and assembly research uses the same task family [16, 24]. Each participant attempts three puzzles in each of three conditions. We ask **RQ1**: How do the conditions differ in correct-piece progress, completion, and round duration? **RQ2**: How do they differ in reported workload and frustration, and how do the assisted policies compare in intervention experience?

Our contributions are:

- (1) A controlled extension of periodic versus physiology-triggered robotic assistance to physical spatial problem solving, with an unassisted comparator and common hints and piece retrieval.
- (2) Participant-level evidence that both assistance policies support correct-piece progress, completion, and reported workload relative to no assistance, without establishing superiority of either assisted policy.
- (3) An assessment of intervention experience and final-survey feedback on hint frequency and robotic piece retrieval.

2 Related Work

2.1 Assistance timing and proactive coordination

Task-level adaptation includes intervention timing and coordination, not only robot motion [13]. Andriella et al. learn proactive assistance from user profiles and memory-game state [1]; PACE estimates action completion from hand movement and reports longer participant waiting under explicit queries than proactive assistance [10]. Ramnauth et al. model assistance utility, recipient costs, relative skill, and task parallelizability, evaluating appropriateness judgments in online vignettes with 215 participants rather than live interventions [21].

Other systems use behavioral or conversational cues. Lavit Nicora et al. analyze assembly gaze with 37 participants and pilot gaze-based initiation with 10 volunteers, treating gaze as readiness for joint action [15]. Tanneberg et al.’s LLM-based Attentive Support decides whether to assist or remain silent using scene and dialogue information; evaluation uses constructed scenarios and a robot demonstration, not a controlled participant comparison [26]. A separate mentalising/Q-learning memory-game study with 56

analyzed participants reports better performance and acceptance, but explanation detail differs between conditions and timing remains fixed [2].

Proactivity need not improve performance. In Vitry et al.’s escape-room study, 56 analyzed participants in 28 pairs interact more with a proactive robot without a significant overall performance difference; both conditions receive scheduled hints [31]. Candon et al.’s 71-participant Space Invaders study finds that reminders before robot behavior changes elicit faster and more immediate feedback, but not a significant whole-game timing effect [6]. Cao et al. study user-initiated interruptions with 21 participants; unsuccessful handling is associated with lower perceived inclusion and discussion satisfaction in exploratory analyses [7]. These studies distinguish initiating assistance, soliciting feedback, and responding to interruptions. Our comparison concerns the first of these.

2.2 Physiological information in assistance

Yang et al.’s surgical system uses EEG and eye tracking to inform adaptive suction, with a periodic comparator selected to approximate earlier observed assistance frequency [33]. Earlier work by Teo et al. uses individualized physiological workload markers to trigger aid during robot supervision, imposing aid later when it has not been triggered [27]. These studies establish approaches to physiological assistance timing across tasks with different demands, sensors, and forms of support.

Other work integrates physiological measurement with broader adaptation strategies. Hostettler et al. adapt robot behavior to user distance while measuring pupil responses; direct pupil-driven adaptation is a future direction [12]. Ojsteršek et al. personalize robot parameters using a preliminary skills test and analyze ECG recordings after the experiment [17]. Korivand et al. develop physiological task-load prediction and Q-learning-based adjustment, while explicitly reporting that their recorded wristband data could not be integrated directly for real-time use [14].

Recent systems extend physiological adaptation beyond robotic task execution. GuideAI combines cardiac, gaze, and behavioral information to adapt learning content, pacing, and feedback [22]. In a single-surgeon simulation study, Wei et al. identify subjective workload and mean heart rate among the influential features in a task-performance prediction model [32]. Their tasks and inference methods differ from our assistance policy.

Prajod et al. examine ECG-derived heart-rate variability and facial-expression estimates under different robot pacing conditions in collaborative assembly. Their perceived-challenge classifier is evaluated offline, and the adaptive condition uses Wizard-of-Oz judgments of task progress rather than physiological triggers [19].

Bhagat Smith and Adams review workload estimation for unknown tasks, emphasizing that the relevance of physiological signals can change across tasks and individuals. They assess machine learning approaches by portability, model complexity, and adaptability, and identify domain generalization and few-shot learning as promising directions requiring further empirical investigation [4].

Pereira et al.’s review documents heterogeneous workload measures and mixed cardiac findings across HRC studies [18]. Capponi et al. similarly find no clear RMSSD pattern across their assembly configurations and distinguish cognitive effort from stress [8]. Cardiac measurement guidelines distinguish heart rate from beat-to-beat variability and emphasize signal quality and the influence of physical activity on wearable measurements [20]. We therefore evaluate HR-rise-triggered assistance through its effects on task performance and participant experience.

2.3 Physical tasks and participant experience

Tabatabaei et al. study gaze around robot failures during collaborative tangram solving [24]. SensCogAR uses tangrams as a proxy for small-object assembly, manipulating the visibility of piece contours to vary task demand [16]. A preliminary three-participant assembly study by Caiazzo et al. compares manual, collaborative, and guided collaborative conditions, with EEG used for workload assessment [5]. MRChaos uses tangrams to investigate robotic reasoning, planning, and manipulation from silhouette targets and extends its assembly approach to cutlery and soda-can arrangements [34]. These task settings combine spatial interpretation, manipulation, and interaction with robot assistance.

Cavicchi et al.’s narrative review of humanoid robots in cognitive-conflict tasks cautions that social cues can distract as well as support performance and recommends delegating parts of the main task to enable cognitive offloading [9].

In an assembly study with 20 participants, van Dijk et al. report lower scores on five workload dimensions under human-led collaboration and lower mental and temporal demand with slower robot pacing. Pacing changes action onset while robot movement speed remains constant [29].

Varrasi et al. compare human and robot guidance in a modified Trail Making Test with 60 younger and older adults. Older adults report greater workload under robot than human assistance, whereas the corresponding difference in younger adults is not significant [30].

In supply-chain operations, Smit et al. model collaborative human-robot order picking and jointly optimize picking efficiency and workload fairness, measured through lifted product mass, in simulation [23].

In a medical-training simulation with 84 participants, Tanjim et al. use a Wizard-of-Oz robotic crash cart to compare speech and light cues for object search and medication reminders. Verbal object-search guidance with visual reminders yields lower reported workload and higher perceived usefulness and ease of use than the reversed cue assignment or no feedback [25]. This comparison concerns assistance format rather than physiological triggering.

Hart describes the six NASA-TLX workload dimensions and the use of an unweighted overall score [11]. We assess these dimensions alongside helpfulness, timing, seamlessness, and frustration relief.

2.4 Wizard-of-Oz evaluation

Wizard-of-Oz methods support the study of robot interactions through human-operated delivery. A recent HRI workshop proposal by Thunberg et al. emphasizes the practical, ethical, and methodological tensions of the wizard’s role [28].

Bejarano et al.’s interviews with six HRI researchers identify challenges involving operator response processing, robot delays, unpredictable participants, and control precision [3]. These findings make operator delivery and interface constraints relevant to interpreting Wizard-of-Oz interactions.

In our setup, the condition determines when assistance is offered: every 30 seconds or when an HR rise is flagged. The researcher delivers a task-relevant hint or robot cue, and the robot operator executes the corresponding program. This arrangement supports the comparison of timing policies using a common assistance interface.

3 Formative Assessment of Tangram Solving

3.1 Assessment procedure and task experience

A formative assessment collected 50 substantive responses: 38 undergraduates, nine graduate students, and three faculty members. Ages ranged from 18 to 30 years ($M = 21.96$, $SD = 3.57$); 25 respondents identified as men, 20 as women, and five preferred not to disclose gender. A Google Form shared through college mailing groups linked to an online square tangram with a timer. Respondents attempted it and reported their experience and expected assistance preferences. Some also participated in the robot experiment; the samples are not treated as independent replications.

Twenty-nine had previously solved tangrams, including four regular solvers; ten had heard of them without solving them and eleven were unfamiliar. Twenty-eight reported complete solutions (56%), 18 partial solutions (36%), and four no solution (8%). Thirty-seven (74%) felt stuck at least once. Twenty-six selected a solving time below one minute, 16 selected one to five minutes, three selected five to ten minutes, and five selected not attempting or solving the puzzle. Completion and time are self-reported categories, not instrumented performance measures.

3.2 Assistance expectations and task rationale

Respondents independently rated hypothetical assistance from 1 (most helpful) to 5 (least helpful), rather than ranking the options. Mean ratings were 2.58 ($SD = 1.34$) for a simpler puzzle, 2.76 ($SD = 1.22$) for a text hint, 2.80 ($SD = 1.14$) for a video tutorial, and 2.94 ($SD = 1.27$) for a robot showing piece placement (Figure 1). Responses requested first-piece hints, visual placement guidance, and help at difficult moments. These descriptive expectations motivate the task and assistance modalities; they do not demonstrate robot efficacy.

Expected helpfulness ($n = 50$; lower = better)

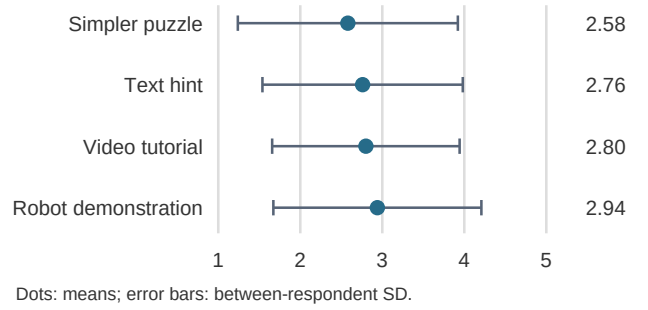


Figure 1: Expected helpfulness in the formative assessment ($N = 50$). Dots show means and bars show SD; lower 1–5 ratings indicate greater expected helpfulness.

Each experimental tangram requires all seven pieces to reproduce a target shape: two large triangles, one medium triangle, two small triangles, a square, and a parallelogram. The seven colored pieces together form a 10×10 -inch square. We use nine colored-piece targets, with solutions available to the researcher.

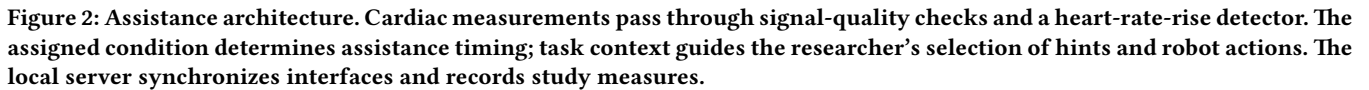
Participants interpret the target, rotate and place pieces, and revise arrangements without a prescribed sequence. Hints address orientation or adjacency; retrieval brings a relevant piece to the workspace. Common materials support repeated comparisons across targets.

4 Study Method: Assistance System and Experimental Protocol

4.1 Architecture and apparatus

Our system coordinates three browser interfaces through a local server (Figure 2). The researcher dashboard displays the puzzle solution, workspace camera view, timer, physiological feedback, and assistance controls. The participant laptop presents instructions, text hints, the timer, and questionnaires. The robot-operator display identifies the piece-specific program to execute. The server synchronizes the interfaces and records task outcomes, interventions, and questionnaire responses.

The setup combines an Orangewood OWL63 robotic arm, a Maxim H Band wrist sensor, and a Logitech C270 workspace camera. An opaque partition separates the participant from two researchers, who administer the task and operate the arm (Figure 3). The camera is mounted on top of the partition and covers the participant table. The participant faces a workspace with a puzzle assembly area in front of and to their left, a drop spot to their right, and a laptop for hints and questionnaires. Unused pieces occupy fixed, labeled pickup positions. The arm retrieves a selected piece using one of seven piece-specific programs and delivers it to the drop spot; the participant places it in the target arrangement. Preset text hints describe piece location, orientation, or adjacency, providing consistent assistance content across participants. Hints are accompanied by an audible notification. The researcher selects either a hint alone or a robot action paired with a hint, based on the current arrangement.



The dashboard displays the flag alongside the workspace view. In the adaptive condition, the flag initiates assistance, while the puzzle arrangement determines the content of the hint or robot action. The flag is a heart-rate-rise cue for assistance timing; participant workload and frustration are assessed through questionnaires.

We measure binary completion, attempt duration, and correct-piece counts recorded in researcher notes and photographs of final arrangements. Counts range from 0 to 7; solved rounds receive 7. Where multiple orientations are admissible, we retain the orientation yielding the highest correct-piece count. Attempt duration includes solved and unsolved puzzles.

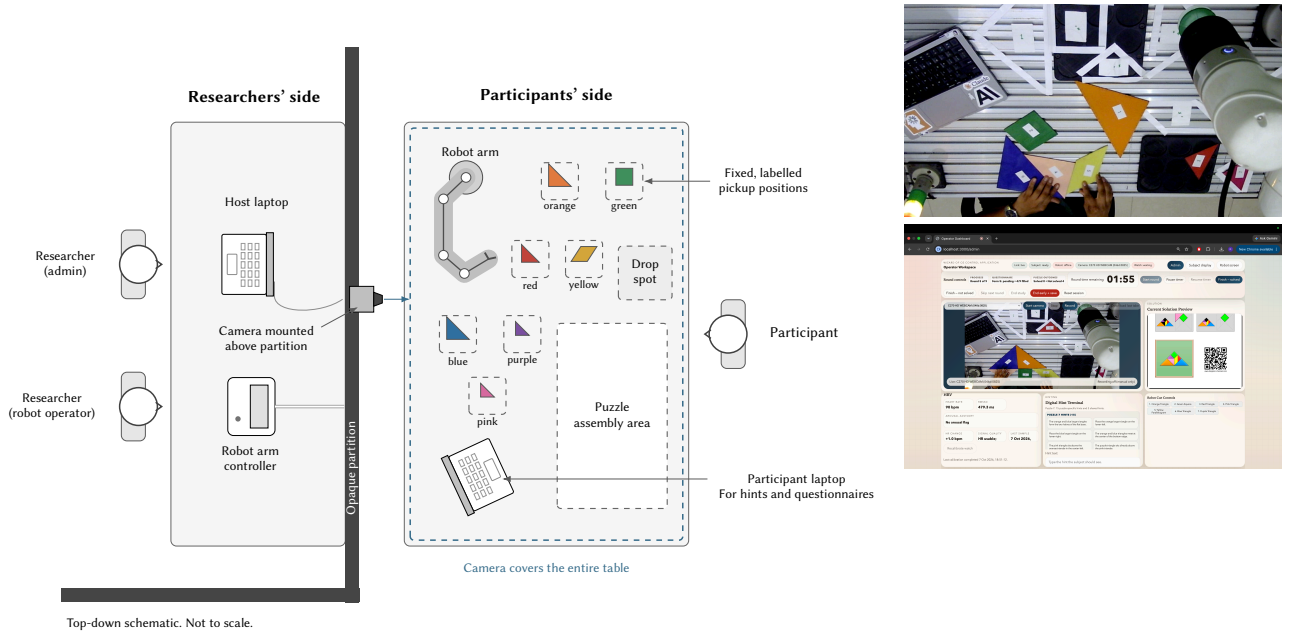


Figure 3: Wizard-of-Oz setup: top-down schematic (left), overhead workspace photograph (upper right), and operator dashboard (lower right). A partition separates researchers from the participant; its camera observes the table, including pickup positions, drop spot, assembly area, and laptop. The dashboard shows an illustrative interface state, not aggregate study results. Schematic not to scale.

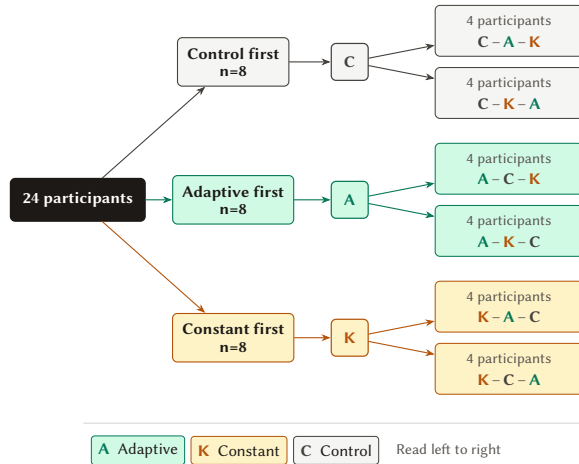


Figure 4: Balanced condition exposure: four participants per order. C denotes control, K constant, and A adaptive; sequences read left to right.

Round-specific questionnaires assess six adapted NASA-TLX items on seven-point scales (1–7): mental, physical, and temporal demand, performance, effort, and frustration. Performance runs from failure to perfect and is reversed as $8 - P$. We average the six aligned burden ratings without weighting and rescale the mean B as $(B - 1)100/6$. This is an adapted unweighted 0–100 workload score, not the original weighted NASA-TLX instrument.

Assisted rounds also assess helpfulness, timing, seamlessness, and frustration relief. Final helpfulness, efficacy, trust, and reliance ratings summarize the whole study, not individual conditions.

All 24 participants contribute task outcomes, durations, and round-specific ratings, with 72 puzzle attempts per condition. We compare conditions using two-sided paired t tests on each participant’s three-round condition averages and report mean differences, 95% confidence intervals, and Cohen’s d_z . To account for multiple comparisons, we apply Holm correction within three groups: completion and overall workload (six comparisons), correct-piece counts (three), and secondary measures (25 across duration, workload dimensions, and assistance ratings). Statistical significance is assessed at adjusted $p < .05$.

5 Findings

5.1 Task performance

Both assisted conditions improved task performance relative to control (Table 1; Figure 5). Across 216 attempts, 90 puzzles were completed: 15.3% in control, 59.7% with constant assistance, and 50.0% with adaptive assistance.

Table 1: Performance by condition, $n = 24$. Pieces and duration show means (SDs) of participant-condition averages; completion shows totals across 72 attempts per condition.

Measure	Control	Constant	Adaptive
Pieces / 7	3.43 (1.40)	5.68 (1.05)	5.43 (1.13)
Completed / 72	11	43	36
Duration (s)	292.4 (22.5)	248.2 (53.3)	265.4 (29.1)

Relative to control, constant and adaptive assistance increased correct-piece counts by 2.25 and 2.00 pieces and completion by 44.4 and 34.7 percentage points, respectively (all Holm-adjusted $p < .001$). Constant had higher observed progress and completion than adaptive, but these policy contrasts were not significant ($p = .273$ and $p = .296$, respectively).

Mean attempt duration was 44.2 seconds shorter under constant assistance and 27.0 seconds shorter under adaptive assistance than control (both secondary-adjusted $p = .006$). The adaptive-constant contrast was not significant ($p = 1.000$). These durations include unsuccessful attempts and occasional overruns beyond the nominal five-minute limit, rather than measuring time to successful completion alone.

5.2 Subjective workload

Adapted workload averaged 62.46 in control, 49.50 with constant assistance, and 46.06 with adaptive assistance (Table 2; Figure 5). Reductions against control were 12.96 points for constant (95% CI [8.52, 17.40], $t(23) = -6.04$, $d_z = -1.23$) and 16.40 for adaptive (CI [10.84, 21.95], $t(23) = -6.10$, $d_z = -1.25$), both core-adjusted $p < .001$. The adaptive-constant difference was not significant ($p = .296$).

Table 2: Workload by condition, mean (SD), $n = 24$. Items use 1–7; performance runs from failure to perfect, while higher demand, effort, and frustration indicate greater burden. Overall workload is rescaled to 0–100.

Measure	Control	Constant	Adaptive
Mental demand	5.71 (0.94)	4.82 (0.98)	4.63 (1.46)
Physical demand	3.65 (1.68)	3.28 (1.37)	3.06 (1.60)
Temporal demand	3.92 (1.38)	3.63 (1.25)	3.56 (1.45)
Performance	2.93 (1.39)	4.54 (1.47)	4.94 (1.23)
Effort	5.44 (0.69)	4.71 (0.95)	4.78 (1.19)
Frustration	4.69 (1.32)	3.93 (1.23)	3.51 (1.06)
Workload / 100	62.46 (11.26)	49.50 (12.93)	46.06 (15.28)

After correction across the 25 secondary tests, both assisted conditions reduced mental demand and increased perceived performance relative to control (all $p < .001$). Constant reduced effort ($p = .004$); adaptive’s effort comparison was not significant ($p = .078$). Adaptive reduced frustration versus control ($p = .002$), whereas constant did not ($p = .086$). Physical and temporal demand comparisons were not significant. No workload dimension differed significantly between assisted policies.

Frustration averaged 4.69 in control, 3.93 in constant, and 3.51 in adaptive. Adaptive minus constant was -0.42 (CI $[-0.90, 0.07]$, $t(23) = -1.77$, adjusted $p = .997$; Figure 6). This contrast did not establish lower frustration under adaptive than constant assistance.

5.3 Intervention experience

Adaptive received descriptively higher means on all four assistance items (Table 3). The timing contrast was $+0.56$ points ($t(23) = 2.91$, raw $p = .008$), but not significant after secondary-family correction ($p = .119$). Helpfulness, seamlessness, and frustration-relief comparisons also did not pass correction (Figure 7). These ratings combine text hints and robot retrieval rather than isolating either modality.

Table 3: Assistance ratings, mean (SD), $n = 24$. All items use 1–7, with higher ratings indicating a better experience.

Measure	Constant	Adaptive
Helpfulness	4.58 (1.26)	4.83 (1.35)
Timing	4.68 (1.48)	5.24 (1.40)
Seamlessness	4.01 (1.39)	4.10 (1.39)
Frustration relief	4.42 (1.35)	4.82 (1.46)

5.4 Overall experience

Fifteen final surveys gave mean helpfulness 4.93 ($SD = 1.62$), efficacy 4.87 ($SD = 1.81$), trust 4.60 ($SD = 1.76$), and agreement with following uncertain robot guidance 4.93 ($SD = 1.67$).

Two final-survey comments illustrate the experience of assistance. One participant wrote, “*The hints were helpful and overall well timed, but the actual robot arm moving the pieces were not adding anything more*”. Another wrote, “*sometimes hints being given too often was counterintuitive and more distracting than better spaced hints, even though any hint helps*.” These comments distinguish the value of guidance from retrieval overhead and the costs of frequent intervention.

6 Discussion, Limitations, and Future Work

6.1 Task progress and the experience of assistance

Both assisted conditions support progress and reported workload relative to control. Constant has the highest observed completion and shortest mean duration; adaptive has better mean timing ratings and lower frustration. However, no corrected assisted-policy contrast establishes superiority. A practical choice between policies therefore remains unresolved, rather than demonstrating that adaptive assistance is preferable or equivalent.

Regular hints may provide repeated opportunities to advance an arrangement, whereas HR-rise-triggered hints may concentrate help at different moments. These are explanations to test, not mechanisms established by the present data. The final-survey comments suggest evaluating content and physical contribution separately: piece retrieval leaves placement to the participant and may add movement overhead, while frequent hints may interrupt reasoning.

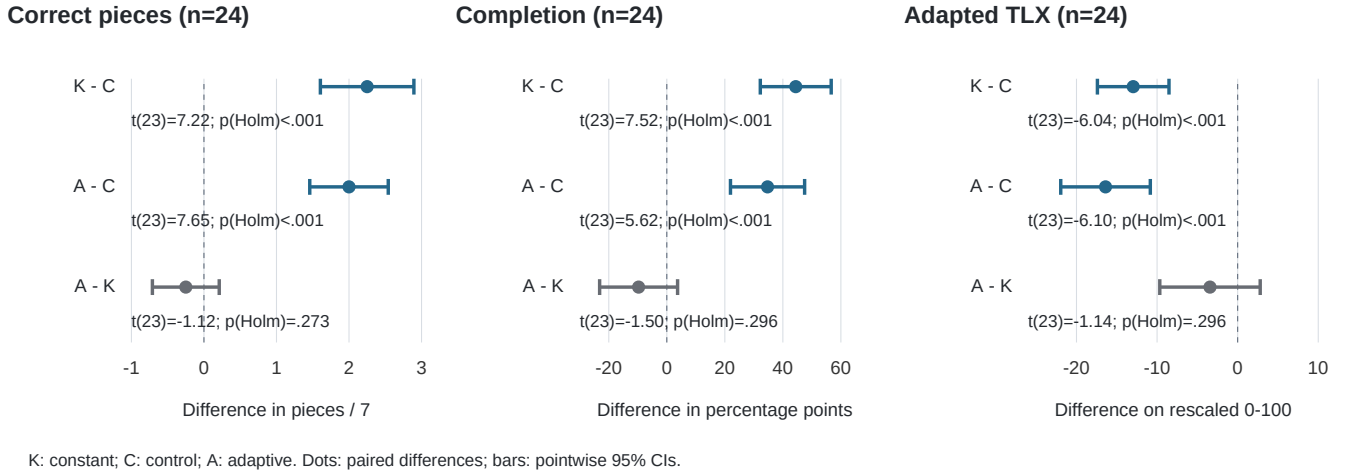


Figure 5: Participant-level paired contrasts for correct pieces, completion, and adapted workload. Dots show mean differences and bars pointwise 95% CIs. Labels give t statistics and Holm-adjusted p values from the respective three-test piece or six-test core family. K: constant; C: control; A: adaptive. Blue denotes adjusted $p < .05$ and gray denotes adjusted $p \geq .05$.

Frustration (n=24; lower is better)

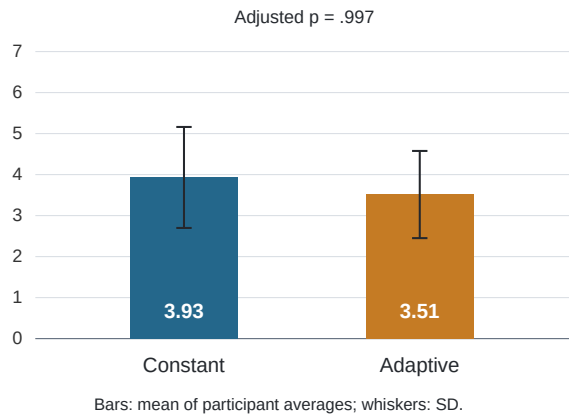


Figure 6: Frustration under constant and adaptive assistance ($n = 24$). Bars show means of participant-condition averages; whiskers show SD. Lower seven-point ratings indicate less frustration. The paired comparison gives adjusted $p = .997$.

6.2 Physiological feedback and limitations

The trigger detects an HR rise, not validated psychological stress. Wearable cardiac measurements are sensitive to movement and sensor contact [20]. The observed assistance benefits therefore do not establish that the HR-rise flag measures stress or workload. Researchers observed some waiting for assistance, but did not systematically code its prevalence or condition distribution. Participants were not told the adaptive trigger rules. Future studies could test explanations and user-requested help, while accounting for PACE’s finding that explicit queries can increase waiting [10].

Assistance ratings (n=24; higher is better)

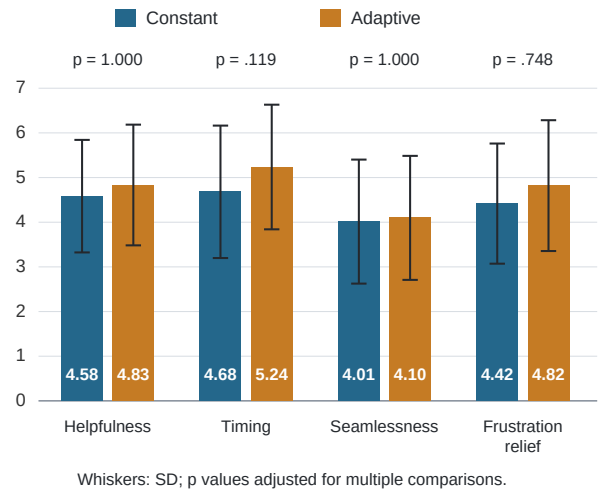


Figure 7: Constant and adaptive assistance ratings ($n = 24$). Bars show means of participant-condition averages; whiskers show SD. Higher seven-point ratings indicate a better experience. Labels show adjusted paired-test p values.

The sample consists of 24 college-recruited volunteers aged 20–27 years ($M = 21.96$), mostly undergraduates with graduate students/researchers and faculty also participating. This age range and recruitment context limit generalization to older adults and experienced industrial workers. Balanced orders do not remove puzzle, period, device, or learning effects from the paired analysis. Frustration is a component of the workload composite, so the two measures are not independent.

Future work should isolate hint delivery from retrieval or direct placement, relate intervention events to observable progress, and use models that account for puzzle and order. Tangrams share orientation and placement demands with assembly [34], while retrieval resembles material provision in order picking [23]. Transfer to either setting requires its own safety, precision, and workflow evaluation.

7 Ethical Considerations

Participation was voluntary and consent was obtained before the task, with the option to pause or stop without penalty. Data comprised cardiac measurements, survey responses, and consented workspace recording/live viewing limited to the table and hands. Participant codes identify analysis records; any release must protect identifying information and respect consent coverage. Concealed Wizard-of-Oz operation was not disclosed during or after the task.

8 Conclusion

We compared unassisted, periodic, and HR-rise-triggered assistance in 24 participants' physical tangram attempts. Both assisted conditions yielded greater correct-piece progress, higher completion, and lower adapted workload than control. Constant had the best observed task means; adaptive had favorable intervention-experience means, without corrected evidence of an assisted-policy difference. The results support assistance in this task while leaving policy choice, physiological validity, and the added value of robot retrieval open.

AI Assistance Disclosure

AI-assisted tools were used in preparing this manuscript. The authors are responsible for the accuracy, originality, and integrity of the work, including its citations.

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